

The elimination method



- 1
- a 7 b $x = -3$ c $y = \frac{3}{7}$
- 2
- a $x = -\frac{16}{3}, y = 3$ b $x = 2, y = -9$
c $x = 4, y = -9$ d $x = 4, y = -\frac{13}{3}$
e $x = -4, y = \frac{23}{3}$ f $x = 0, y = 3$
g $x = -4, y = -\frac{13}{9}$ h $x = 2, y = \frac{11}{7}$
i $x = -2, y = -2$ j $x = -20, y = \frac{53}{3}$
k $x = 3, y = -5$ l $x = -5, y = 7$
m $x = -\frac{2}{3}, y = \frac{1}{3}$ n $x = -\frac{32}{7}, y = -3$
o $x = -\frac{7}{9}, y = -2$ p $x = 1, y = -3$
q $x = 5, y = -2$ r $x = 3, y = 1$
- 3
- a
- i $3x + 11y = -48$ and $-6x + 21y = -76$
ii $x = -\frac{4}{3}$ and $y = -4$
- b
- i $2x + 3y = 5$ and $5x + 4y = 2$
ii $x = -2$ and $y = 3$
- c
- i $4x + 3y = 20$ and $8x - 3y = 4$
ii $x = 2, y = 4$
- d
- i $6x + 5y = 240$ and $x + 3y = 66$
ii $x = 30, y = 12$
- 4
- a $x = 2, y = 8$ b $x = 2, y = -9$ c $x = 8, y = 3$ d $x = 5, y = 5$
e $x = -9, y = 4$ f $x = 6, y = 7$ g $x = -20, y = 15$ h $x = 3, y = \frac{23}{3}$
i $x = -1, y = -1$ j $x = \frac{1}{2}, y = \frac{3}{2}$ k $p = -\frac{3}{2}, q = \frac{23}{10}$ l $x = -2, y = 3$



Applications

5

a $x + y = 122$

b $x - y = 12$

c $x = 67$

d $y = 55$

6

a $9x + 2y = 26$

b $3x + 4y = 22$

c $x = \$2$

d $y = \$4$

7

a $x = 1$

b $y = 68$

c 2016

d 68%

8

a $2x + y = 27, 5x - y = 43$

b $x = 10, y = 7$

9

a $x + y = 10, 6x - 5y = 5$

b $x = 5, y = 5$

10

a $7x + y = 64, 3x - y = 16$

b $x = 8, y = 8$

11

a $y = 10x, y = 7x + 18$

b Son is 6 and mother is 60 years old.